

The Transport Trait

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Revision: 1 Last modified: 2026-06-25T00:00:00Z

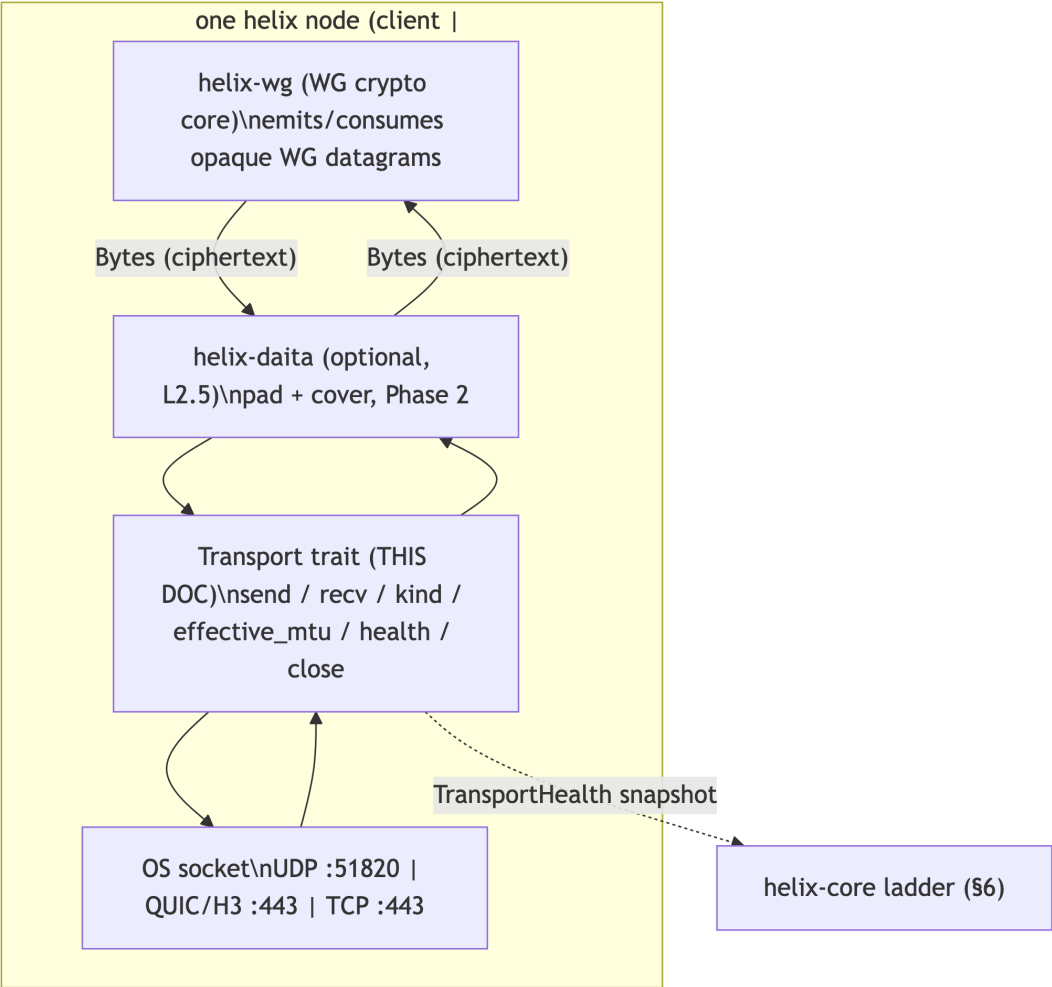
Volume 2 (Data Plane) nano-detail specification — deepens §3 ("The Transport trait") of the pass-1 data-plane overview [01-data-plane §3]. SPEC ONLY: describe **what to build** (signatures, wire formats, state machines, error taxonomy, edge cases, security, performance budget, test points); do **not** write the shipping product. Source evidence cited inline by id — [01-DP §N] (final/01-data-plane.md), [04_ARCH §N] (04_VPN_CLD/HelixVPN-Architecture-Refined.md), [04_P0 §N] (04_VPN_CLD/HelixVPN-Phase0-Spike.md), [04_P2 §N] (04_VPN_CLD/HelixVPN-Phase2-Parity.md), [research-masque] (v09-research/research-masque.md), [SYNTHESIS §N] (v09-research/_SYNTHESIS.md). Anything not grounded in that evidence base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

0. Position, ownership, and the one-sentence contract

helix-transport is the data plane's **L2 carrier crate**: a decoupled, project-agnostic Rust crate (vasic-digital/helix_core workspace, snake_case, flat submodule per §11.4.28/.29/.74) [01-DP §2, SYNTHESIS §9]. It owns exactly one public abstraction — the Transport trait — plus its config enum, its constructor (dial), its error taxonomy, and the concrete impls (plain_udp, masque, connect_ip, shadowsocks, udp_over_tcp, lwo, hysteria2) [01-DP §2, §3].

The one-sentence contract. A Transport is a *bidirectional carrier for already-encrypted, unreliable WireGuard datagrams between this node and exactly one peer endpoint* — it never sees plaintext (invariant I1), never reorders or retransmits (invariant I2), and is the *same code on client, connector, and edge* (invariant I4) [01-DP §0.1, §3, 04_P0 §4.2].

This document does **not** own: the WG crypto core (helix-wg, §4 of [01-DP]), the orchestrator three-loop (helix-core, [01-DP §5]), the WatchNetworkMap wire contract that *sources* TransportConfig (doc 03), nor the per-platform TUN shims (doc 06). It owns the seam between helix-wg above and the OS socket below.



1. Non-negotiable transport invariants (the contract every impl must keep)

These bind **every** Transport impl. A violation is a release blocker (§11.4 anti-bluff at the data-plane layer) [01-DP §0.1].

#	Invariant	Mechanical consequence in this trait	Source
I1	The transport never sees plaintext; it carries only already-	send/recv operate on opaque Bytes; no impl ever parses the WG header beyond what an obfuscation layer mangles	[04_P0 §4.2]

#	Invariant	Mechanical consequence in this trait	Source
	encrypted WG datagrams.	(lwo), and even then it mutates ciphertext bytes, never decrypts.	
I2	The transport carries unreliable datagrams , never an ordered byte stream; WG's own loss/reorder semantics are preserved; no head-of-line blocking.	MASQUE uses QUIC DATAGRAM frames (RFC 9221), not a QUIC stream. TCP-based carriers (shadowsocks, udp-over-tcp) accept the HoL penalty <i>only</i> as last resort and frame each datagram independently so a parser can still emit per-datagram.	[04_P0 §4.2 design note], [research-masque §1]
I3	Control plane is never in the packet path; transports keep forwarding if control is down (fail-static).	A Transport, once dialed, holds no live dependency on the coordinator — it only needs its TransportConfig snapshot (resolved at dial).	[01-DP §0.1]
I4	One transport crate, three consumers.	The trait + impls compile identically into helix-client, helix-connector, helix-edge; the edge's de-obfuscation is the inverse of the same encode_*/decode_* functions the client uses.	[04_ARCH §3.2, §5.5]
I5	No-logging by construction.	Transport impls hold only aggregate counters in TransportHealth; no per-packet, per-connection durable state is written. Tracing is opt-in, counters-only, never payload.	[01-DP §0.1]
DG	Datagram-message boundary preservation. One send(dg) ⇒ exactly one recv() on the peer yields that same dg's bytes (modulo loss). No coalescing, no splitting across recv calls.	UDP/QUIC-DATAGRAM are message-oriented natively; TCP carriers MUST re-impose the boundary with an explicit length prefix (§4.3, §4.4).	derived from I2

Honest boundary (§11.4.6). I2 is preserved *over UDP and QUIC-DATAGRAM natively*. Over TCP (shadowsocks, udp-over-tcp) the wire is an ordered stream; the impl re-creates datagram boundaries with a length prefix but **cannot** restore loss-tolerance — a lost/late TCP segment HoL-blocks every following datagram. This is an accepted, documented degradation, not a bug; the ladder (§6) prefers TCP carriers last for exactly this reason [01-DP §3.5, §3.6].

2. The Transport trait — full signature & per-method nano-contract

The pass-1 trait [01-DP §3.1] is the frozen Phase-0 surface [04_P0 §4.2, §14]. This section pins the exact semantics of every method: argument ownership, cancel-safety, blocking behaviour, and the closed set of errors each may return.

```
// helix-transport/src/lib.rs
use async_trait::async_trait;
use bytes::Bytes;
use std::net::SocketAddr;
use std::time::Duration;

/// A bidirectional carrier for ALREADY-ENCRYPTED WireGuard
/// datagrams.
/// Invariants: I1 (never plaintext), I2 (unreliable datagrams,
/// never a stream),
/// DG (1 send => 1 recv, boundary-preserving). All async methods
/// are cancel-safe:
/// dropping the returned future at any await point leaves the
/// Transport usable.
#[async_trait]
pub trait Transport: Send + Sync {
    /// Send exactly one WG datagram toward the single peer
    /// endpoint.
    /// Ownership: takes `Bytes` by value (cheap ref-counted clone
    /// upstream).
    /// Returns Ok(()) once the datagram is handed to the OS /
    /// QUIC stack
    /// (NOT once the peer acks – there is no ack; this is fire-
    /// and-forget, I2).
    /// Errors: Oversize (dg.len() > effective_mtu+headroom),
    /// Closed, Io, Quic,
    /// EndpointBlocked (ICMP port-unreachable /
    /// connection refused seen).
    async fn send(&self, datagram: Bytes) -> Result<(),
        TransportError>;

    /// Receive the next WG datagram from the peer.
    /// Cancel-safe: usable as a branch in `tokio::select!`; a
    /// cancelled recv
```

```

/// MUST NOT drop an already-dequeued datagram (no data loss
on cancel).
/// Blocks (async) until one full datagram is available or the
carrier dies.
/// Errors: Closed (graceful peer FIN / QUIC
CONNECTION_CLOSE), Io, Quic.
async fn recv(&self) -> Result<Bytes, TransportError>;

/// Stable, lowercase, hyphenated label for logs/metrics/
status:
/// "plain-udp" | "masque-h3" | "connect-ip" | "shadowsocks"
/// | "udp-over-tcp" | "lwo" | "hysteria2". MUST match
`TransportKind` (§3.3).
fn kind(&self) -> &'static str;

/// MTU (bytes) the upper layer (WG) may use over this
transport, AFTER
/// per-transport overhead (§7). Constant for the lifetime of
the impl
/// UNLESS path-MTU discovery lowers it (then it MAY shrink,
never grow
/// above the dialed ceiling). Always <= 1420 and >= 1280
(IPv6 floor).
fn effective_mtu(&self) -> u16;

/// Cheap, lock-light snapshot of liveness for the ladder's
decisions (§6).
/// MUST be callable at >= 10 Hz without contending the hot
send/recv path.
fn health(&self) -> TransportHealth;

/// Graceful close: flush QUIC / send TCP FIN / drop the UDP
socket.
/// IDEMPOTENT – a second call is a no-op Ok(()). After close,
send/recv
/// return Err(Closed). MUST complete within CLOSE_GRACE
(§5.6).
async fn close(&self) -> Result<(), TransportError>;
}

```

2.1 Per-method nano-contract table

Method	Concurrency	Cancel-safe?	Returns Ok when	Hot-path?	May trigger ladder escalation?
send	callable concurrently with recv from another task; MUST be &self (interior mutability)	yes — partial send is impossible at the datagram level (UDP/	datagram handed to OS/ QUIC tx queue	yes	only via repeated EndpointBlocked/ Io raising send_errors, surfaced through health() to §6

Method	Concurrency	Cancel-safe?	Returns Ok when	Hot-path?	May trigger ladder escalation?
		QUIC-DATAGRAM are atomic; TCP send is buffered & resumable)			
recv	one logical reader; MUST NOT be called from two tasks simultaneously (UNVERIFIED: impls MAY internally serialize, but the orchestrator runs a single recv loop per transport [01-DP §5.1])	yes — a cancelled recv re-queues any partially-read state	one whole datagram dequeued	yes	no (recv errors map to Down/Reconnecting at orchestrator, §5)
kind	lock-free, const	n/a	always	metrics only	no
effective_mtu	lock-free, near-const	n/a	always	per-config only	no
health	lock-light snapshot of atomics	n/a	always	sampled, not hot	feeds §6 decisions
close	idempotent	n/a (not awaited under select normally)	carrier flushed/closed	teardown	no

2.2 Object-safety, sizing, and dynamic dispatch

- Transport is **object-safe**: dial returns `Box<dyn Transport>` [01-DP §3.1]. No generic methods, no Self-returning methods, no associated consts in the public surface — all preserved so dyn dispatch works.

- The orchestrator holds the active carrier as `Box<dyn Transport>` and swaps it atomically on a ladder escalation (§6.4) — a single `Box` pointer write under the orchestrator's transport lock, never mid-datagram.
- `Send + Sync` are **required** supertraits: the carrier is shared between the send loop and recv loop tasks (`Arc<dyn Transport>` inside the orchestrator) [01-DP §5.1]. UNVERIFIED: whether the orchestrator wraps in `Arc` vs moves the `Box` into a single combined loop is an orchestrator decision (doc owns §5.1), not a trait constraint.

2.3 TransportHealth — the liveness snapshot

```
#[derive(Clone, Debug)]
pub struct TransportHealth {
    /// EWMA of RTT in ms; None until the first RTT sample lands.
    /// RTT source: per-transport – UDP/lwo: WG keepalive echo
    /// timing
    /// (UNVERIFIED: helix-wg owns keepalive RTT; transport MAY
    /// expose a
    /// carrier-level probe instead); MASQUE/hysteria2: quinn's
    /// path RTT
    /// estimate (quinn exposes `Connection::rtt()`); TCP
    /// carriers: socket RTT
    /// via TCP_INFO where available, else None.
    pub rtt_ewma_ms: Option<u32>,
    /// Wall-clock ms since the last successful recv() of a
    /// datagram.
    /// The ladder uses this for "carrier went silent" detection
    /// (§6.3).
    pub last_recv_age_ms: u64,
    /// Monotonic count of send() errors since dial (Io/
    /// EndpointBlocked/Quic).
    /// Reset only on a fresh dial(), never decremented (I5:
    /// aggregate counter).
    pub send_errors: u32,
}
```

EWMA recommendation (UNVERIFIED, calibrate per §11.4.107(13) on real fixtures, not literature-hardcoded): $\text{rtt_ewma} = 7/8 * \text{rtt_ewma} + 1/8 * \text{sample}$ (the classic TCP SRTT $\alpha=1/8$ [RFC 6298] is the documented starting point; final α set by Phase-0 BENCH evidence).

3. TransportConfig, supporting types, and the wire-config seam

3.1 The config enum (one variant per L2 carrier)

TransportConfig is the **resolved** per-carrier dial parameters. Every variant is hydrated from the NetworkMap pushed by the coordinator (doc 03) — endpoints, SNI, PSKs, session keys all arrive *already policy-filtered* (need-to-know, I6) [01-DP §3.1, §6.2]. The data plane never invents an endpoint or a secret.

```
// helix-transport/src/lib.rs
#[derive(Clone, Debug)]
pub enum TransportConfig {
    /// Baseline UDP carrier; lowest latency/CPU; throughput
    /// baseline (§7).
    PlainUdp      { peer: SocketAddr, bind: SocketAddr },

    /// CONNECT-UDP over HTTP/3 (RFC 9298/9297/9221) – the
    /// headline obfs (§4.2).
    MasqueH3      { url: String, sni: String, bind: SocketAddr,
                    congestion: Congestion },

    /// IP-over-HTTP/3 (RFC 9484). Advanced, feature-gated, Phase
    /// 2+ (§4.5).
    ConnectIp     { url: String, sni: String, bind: SocketAddr },

    /// WG-in-Shadowsocks AEAD over TCP (China-grade DPI) (§4.3).
    Shadowsocks   { peer: SocketAddr, method: SsMethod, psk:
                    SecretBytes },

    /// WG datagrams length-prefixed over one TCP conn; last
    /// resort (§4.4).
    UdpOverTcp    { peer: SocketAddr, tls_sni: Option<String> },

    /// Lightweight per-session keyed header obfs + padding
    /// (§4.6).
    Lwo           { peer: SocketAddr, bind: SocketAddr,
                    session_key: SecretBytes },

    /// Hysteria2 + Salamander QUIC-obfs carrier; feature-gated;
    /// decision D1.
    #[cfg(feature = "hysteria2")]
    Hysteria2     { url: String, sni: String, salamander_pw:
                    SecretBytes },
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)] pub enum Congestion
{ Cubic, Bbr }

#[derive(Clone, Copy, Debug, PartialEq, Eq)] pub enum SsMethod {
    Chacha20Poly1305, Aes256Gcm }
```


Field semantics:

Field	Type	Meaning	Constraint
peer	SocketAddr	the gateway/ connector L1 endpoint to send datagrams to	v4 or v6; resolved, never a hostname
bind	SocketAddr	local socket bind addr (0.0.0.0:0 / [::]:0 for ephemeral)	UDP carriers only
url	String	full HTTPS URL of the gateway H3 listener, e.g. https:// gw.example:443	scheme MUST be https; informs the RFC 9298 URI template (\$4.2.1)
sni	String	TLS SNI presented in the QUIC Initial / ClientHello	drives the fingerprint- mimicry research item (\$8.3)
congestion	Congestion	quinn CC: Cubic default, Bbr for lossy mobile	[01-DP §3.3]
method	SsMethod	Shadowsocks AEAD cipher	reuse shadowsocks- rust primitives [04_P2 §1.1]
psk / session_key / salamander_pw	SecretBytes	per-transport secret, separate from WG keys	§3.2; zeroized on drop
tls_sni	Option<String>	if Some, wrap the TCP carrier in TLS to look like HTTPS	optional camouflage [01- DP §3.6]

3.2 SecretBytes — the secret-handling type (§11.4.10 binding)

SecretBytes is the closed type for every transport secret. It is **distinct from WG keys** — a transport-layer password/PSK, never the device's WG private key (which never leaves the device, doc 02) [01-DP §3.5, §4].

```
// helix-transport/src/secret.rs
/// A transport secret (SS PSK, lwo session key, salamander
    password).
/// - Debug/Display REDACT (prints "SecretBytes(<redacted>)" -
    §11.4.10 no-leak.
/// - Zeroize on Drop (memory scrubbed) - uses `zeroize` crate.
/// - No Serialize/Deserialize derive: secrets arrive via the
    doc-03 wire path,
/// never serialized into a tracked artifact (§11.4.10 /
    §11.4.30).
pub struct SecretBytes(/* zeroize::Zeroizing<Vec<u8>> */);

impl std::fmt::Debug for SecretBytes {
    fn fmt(&self, f: &mut std::fmt::Formatter<'_,>) ->
        std::fmt::Result {
        f.write_str("SecretBytes(<redacted>)" // never the
            bytes (§11.4.10)
        }
    }
}
```

Security rules (release-blocking, §11.4.10): (1) SecretBytes MUST impl Drop zeroizing the buffer; (2) Debug/Display MUST redact; (3) no derive(Serialize); (4) no logging API ever accepts SecretBytes. The SEC test type (§9) plants a secret value and asserts a repo-wide + log-capture grep finds it nowhere.

3.3 TransportKind — the carrier identifier the ladder orders

TransportKind is the lightweight, secret-free enum the ladder (§6) sorts and the coordinator pushes in TransportPolicy.order [01-DP §5.3]. One-to-one with TransportConfig variants and with kind() strings.

```
// helix-transport/src/lib.rs
#[derive(Clone, Copy, Debug, PartialEq, Eq, Hash)]
pub enum TransportKind {
    PlainUdp, MasqueH3, ConnectIp, Shadowsocks, UdpOverTcp, Lwo,
    #[cfg(feature = "hysteria2")] Hysteria2,
}

impl TransportKind {
    pub const fn label(self) -> &'static str { /* matches kind()
        strings (§2) */ }
}
```

4. Per-carrier dial detail & wire formats (byte layouts)

Each subsection pins the on-wire byte layout the impl must produce/consume so the *client encode* and *edge decode* (I4) are provably inverse. WG datagram = the opaque ciphertext payload from `helix-wg`; its own header layout (message type, sender/receiver index, counter, AEAD tag) is owned by WG and treated as opaque here [01-DP §4].

4.1 `plain-udp` — baseline (Phase 0, always available)

Wire = the WG datagram, raw, in one UDP payload. No framing added.

```
L1 IP | UDP(:51820) | [ ---- WG datagram (opaque ciphertext)
---- ]
```

`effective_mtu() = 1420` (`1500 - 20 IPv4 - 8 UDP - ~52 WG hdr+tag` headroom per [04_P0 §4.3]; `1440` over IPv6 path-MTU UNVERIFIED — measured in Phase 0 BENCH, the table value `1420` is the conservative IPv4 default) [01-DP §10]. `dial = bind the UDP socket, connect(peer)` (so `send/recv` need no `addr`), return immediately — there is **no carrier handshake** (the WG handshake rides on top, owned by `helix-wg`). Fastest dial (sub-ms).

4.2 `masque-h3` — CONNECT-UDP over HTTP/3 (the headline obfuscation)

The "QUIC mode" — Mullvad's literal mechanism, WG-over-MASQUE [04_ARCH §0, `research-masque` §2]. Four-RFC layering [`research-masque` §1]:

```
WG datagram (Bytes, opaque ciphertext)
└─ HTTP Datagram (RFC 9297): [ Quarter-Stream-ID (varint) |
Context-ID=0 (varint) | WG payload ]
    └─ QUIC DATAGRAM frame (RFC 9221, UNRELIABLE — matches WG
loss semantics, I2)
        └─ QUIC / HTTP-3 connection to https://gw:443 (looks
like a browser doing H3)
            └─ EDGE: CONNECT-UDP proxy (RFC 9298) strips
framing → WG datagram → kernel WG
```

4.2.1 CONNECT-UDP establishment (once, at dial)

Per RFC 9298 + RFC 9220 Extended CONNECT [`research-masque` §1]:

1. Open a QUIC/H3 connection to `url` (`quinn + h3`), TLS SNI = `sni`, ALPN = `h3`. Negotiate `max_datagram_frame_size` transport param `> 0` (RFC 9221) — if the peer does not offer QUIC datagrams, `dial` fails `HandshakeFailed`

(the carrier is unusable without unreliable datagrams, I2) [research-masque §1].

2. Issue an **Extended CONNECT** request stream: `:method = CONNECT, :protocol = connect-udp, :scheme = https, :authority = <host:port>, :path = /.well-known/masque/udp/{target_host}/{target_port}/` (RFC 9298 URI template; {target} = the gateway's local WG endpoint) [research-masque §1].
3. On 2xx the flow is established; record the **Quarter-Stream-ID** (= the request stream id / 4) used to tag every datagram to this flow (RFC 9297). This is `flow_ctx` in the pass-1 sketch [01-DP §3.3].
4. Thereafter WG datagrams ride as HTTP Datagrams with **no per-packet HTTP round trip** [01-DP §3.3, 04_P0 §5.1].

Maturity caveat (release-relevant, not a spec defect). There is **NO turnkey Rust crate** for RFC 9298 CONNECT-UDP; it is hand-rolled on quinn

- `h3 + h3-datagram` (v0.0.2, pre-1.0) [research-masque §3]. The impl **MUST** hand-write: the Extended-CONNECT request/response, the Quarter-Stream-ID datagram mux, and (edge side) the UDP-socket relay. Mullvad's published `quinn_udp` dependency confirms this is the trodden path [research-masque §2]. Budget as real engineering; this immaturity is the input to decision D5 (edge language, [01-DP §11.2]).

4.2.2 HTTP-Datagram byte layout (the per-packet hot path)

HTTP/3 Datagram (RFC 9297) payload, carried in one QUIC DATAGRAM frame (RFC 9221):

Quarter Stream ID (QUIC	Context ID (QUIC varint)	
Proxied UDP payload		
varint, identifies the	= 0x00 for the raw UDP	= the
WG datagram		
CONNECT-UDP request stream)	payload (RFC 9298 §5)	
(opaque, I1)		

`encode_http_datagram(flow_ctx, wg)` prepends the two varints; `send_datagram` hands it to quinn [01-DP §3.3]. `decode_http_datagram(dg)` strips them back to the WG bytes. Edge = the exact inverse (I4). Context-ID 0 = raw UDP payload per RFC 9298 §5 [research-masque §1].

`effective_mtu()` = **1280** (measure & tune): QUIC long/short header + DATAGRAM frame header + the two varints eat headroom; 1280 is the IPv6 minimum-MTU floor and a safe default [01-DP §3.3, §10]. The exact penalty is **quantified in Phase-0 BENCH** (record MTU/throughput/CPU vs plain-udp) so the ladder cost model is real [04_P0 §5.3].

4.2.3 Masquerade (decoy on non-CONNECT-UDP probes)

The edge :443 listener serves a believable decoy site to anything that is **not** a valid CONNECT-UDP flow (scanners/probes) — native edge behaviour, not Nginx camouflage [01-DP §3.3, 04_P0 §5.4]. Verified by the SEC/IT G2 check: tshark classifies the flow as HTTP/3 with **no WG signature** [04_P0 §8].

4.3 shadowsocks — WG-in-Shadowsocks AEAD over TCP

WG datagrams wrapped in a Shadowsocks AEAD stream — "looks like random/TLS-ish TCP" where even QUIC is throttled [01-DP §3.5, 04_P2 §1.1]. Reuse shadowsocks-rust crypto primitives, do not re-implement AEAD framing [04_P2 §1.1].

Per-datagram wire (re-imposes the DG boundary over the TCP stream):

TCP stream (after SS AEAD session established) carries repeated frames:

[2-byte length prefix (big-endian, = WG datagram len) | WG datagram] → AEAD-encrypt each chunk
recv: AEAD-decrypt → read 2-byte len → read len bytes → one WG datagram (DG boundary restored)

dial = TCP connect to peer, derive the SS session subkey from psk + salt (per SS AEAD spec, owned by shadowsocks-rust), no extra app handshake. `effective_mtu()` ≈ **1380** (TCP MSS – 2-byte length prefix – AEAD tag) [01-DP §10]. Carries the I2 HoL caveat (TCP); ladder rung above QUIC.

4.4 udp-over-tcp — last resort (all UDP blocked)

WG datagrams length-prefixed over one TCP connection — matches Mullvad's `udp2tcp` [01-DP §3.6, 04_P2 §1.2]. Optional TLS wrap (`tls_sni`) to look like HTTPS. Lowest ladder rung; exists only to keep a tunnel *possible*.

TCP (optionally inside TLS if `tls_sni=Some`) stream of frames:
[2-byte length prefix (big-endian) | WG datagram] [2-byte len | WG datagram] ...

`effective_mtu()` ≈ **1380** (TCP MSS – length prefix) [01-DP §10]. Full HoL penalty accepted by design.

4.5 connect-ip — IP-over-HTTP/3 (RFC 9484, advanced, feature-gated)

Proxies **IP packets** (not inner WG) over HTTP/3 via CONNECT-IP [01-DP §3.4, research-masque §1]. **Spec stance:** Phase 2+, behind a Cargo feature; default path stays inner-WG-over-masque-h3. CONNECT-IP **loses I1's "transport never sees plaintext"** at the gateway (the gateway terminates

IP), so it is **not** offered to privacy clients by default — only explicit site-to-site routing where the gateway is already a trusted router [01-DP §3.4].
effective_mtu() $\approx$ 1300 (IP-over-H3, no inner-WG header but QUIC overhead) [01-DP §10].

4.6 lwo — lightweight WG obfuscation (cheap evasion)

Per-session keyed obfuscation of the WG header bytes DPI signatures key on (message-type / reserved fields) + randomized padding — near-zero-cost evasion of *naïve* WG fingerprinting without QUIC overhead [01-DP §3.7, 04_P2 §1.3]. Mullvad's "LWO" is the design reference. Phased: Phase-1 basic XOR/padding, Phase-2 hardened per-session keyed scheme [01-DP §3.7].

Over UDP, per datagram:

```
obfuscate(session_key, wg) = [ mangle WG header bytes via  
keystream(session_key) | WG body | random pad (len  $\in$  [0, P]) ]  
deobfuscate inverts the mangle + strips pad (pad length carried  
in a keyed field – UNVERIFIED exact field; Phase-1 design owns it)
```

I1 preserved: lwo mutates **ciphertext** bytes; it never decrypts. First (cheap) escalation rung above plain-udp [01-DP §3.7]. effective_mtu() $\approx$ 1400 (plain-udp minus padding budget) [01-DP §10].

4.7 hysteria2 — decision D1 carrier (feature-gated)

WG datagrams over Hysteria2's QUIC-datagram channel + Salamander packet obfs; **reuses the same quinn dependency** as masque-h3; the obfs differs, the WG-datagram contract is identical (I2) [01-DP §3.8]. **D1 recommendation: masque-h3 is primary** (literal Mullvad parity, single Rust impl, keeps WG the crypto core); hysteria2 ships as an *additional* Transport impl behind a Cargo feature — a ladder rung and a Phase-0 unblock if MASQUE-in-Rust proves painful [01-DP §3.8, SYNTHESIS §3 D1].
DECISION D1 is SURFACED, not silently resolved (§11.4.66).

4.8 MTU & overhead summary (effective_mtu() per carrier)

Carrier	effective_mtu()	Basis	Source
plain-udp	1420	1500 – 20 IP – 8 UDP – ~52 WG	[01-DP §10, 04_P0 §4.3]
masque-h3	1280 (measure & tune)	QUIC + HTTP- Datagram + 2 varints; IPv6 floor	[01-DP §10, 04_P0 §5.2]
connect-ip	~1300	IP-over-H3, QUIC overhead	derived [01- DP §10]
shadowsocks	~1380		

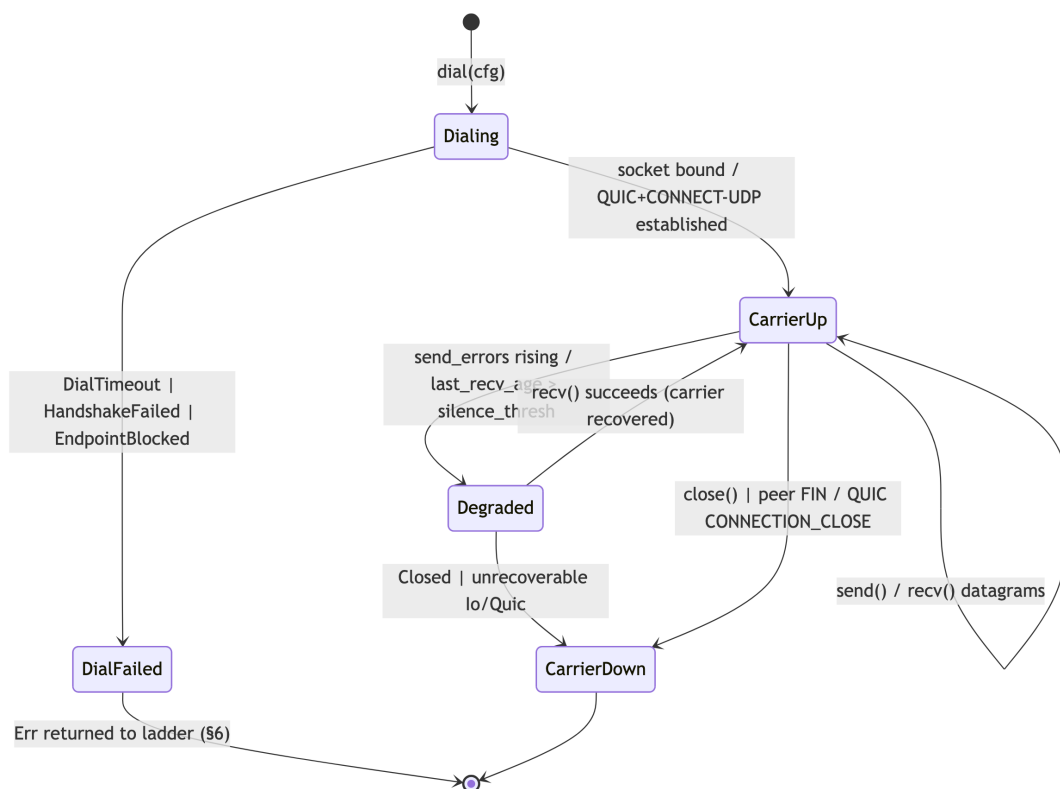
Carrier	effective_mtu()	Basis	Source
		TCP MSS – 2B len – AEAD tag	derived [04_P2 §1.1]
udp-over-tcp	~1380	TCP MSS – len prefix	derived [04_P2 §1.2]
lwo	~1400	plain-udp – pad budget	derived [04_P2 §1.3]
hysteria2	~1280	QUIC + Salamander, same family as masque	UNVERIFIED (measure)

Rules [01-DP §10]: (1) inner WG MTU = `min(active.effective_mtu(), path-MTU-discovered)`; (2) DAITA padding is added **after** WG encrypt and counts against the transport budget, not the inner MTU; (3) Oversize is a **hard error** — the orchestrator lowers inner MTU / L3-fragments rather than silently truncate.

5. Lifecycle state machine (connect / handshake / up / reconnect / down)

A Transport's lifecycle has two coupled layers: the **carrier** layer (this crate — socket/QUIC/TCP up-or-down) and the **tunnel** layer (`helix-wg` handshake on top, owned by [01-DP §4]). The orchestrator's `TunnelStatus` [01-DP §5.2] is the *user-facing* projection of both.

5.1 Carrier-level states (owned by a single Transport impl)



- **Dialing**: `dial(cfg)` runs within a bounded `DIAL_TIMEOUT` (§5.5). For `plain-udp` this is a socket bind (sub-ms); for `masque-h3` it is the full QUIC + Extended-CONNECT establishment (§4.2.1).
- **CarrierUp**: datagrams flow. No WG state here — the carrier is oblivious to whether the WG handshake on top has completed (I1).
- **Degraded**: a *soft* state surfaced via `health()` (rising `send_errors` / growing `last_rcv_age_ms`), not a hard error. The orchestrator/ladder reads it to pre-emptively escalate before a hard failure (§6.3).
- **CarrierDown**: terminal; `send/rcv` return `Err(Closed)`. Reached by `close()` (graceful) or an unrecoverable error.

5.2 Tunnel-level states (orchestrator projection — the UX seam)

The orchestrator composes carrier state with the WG handshake into `TunnelStatus`, broadcast to FFI + metrics + ladder via `tokio::sync::broadcast` [01-DP §5.2]:

```
// helix-core/src/status.rs (consumed here; OWNED by [01-DP §5.2])
#[derive(Clone, Debug)]
pub enum TunnelStatus {
    Connecting, // dial in
    flight(carrier Dialing)
```

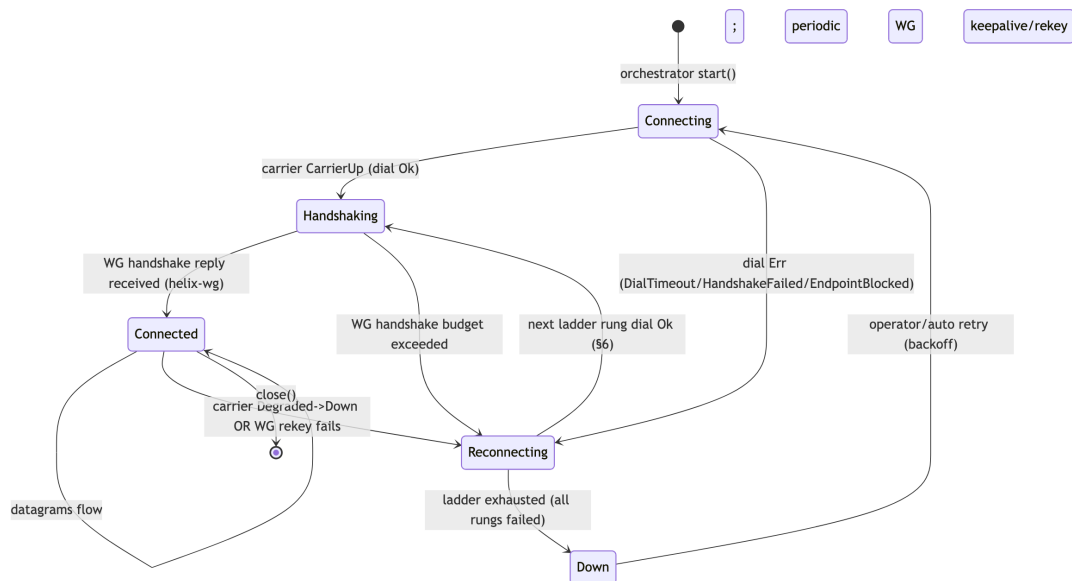


```

Handshaking,                                     // carrier up,
    WG handshake in flight
Connected { transport: String, rtt_ms:
    u32 }, // carrier up + WG established
Reconnecting,                                     // carrier
    down/escalating; ladder advancing
Down { reason: String },                         // all rungs
    exhausted / fatal
}

```

5.3 Combined lifecycle (carrier $\rightleftharpoons$ WG $\rightleftharpoons$ status)



Boundary note (§11.4.6). `Connected{rtt_ms}` is emitted only after the WG handshake reply lands (a `helix-wg` event), not merely on `CarrierUp` — a carrier can be up while the WG tunnel never establishes (e.g. wrong WG peer key). The status enum reflects *tunnel* liveness, the truth the user cares about [01-DP §5.2].

5.4 Reconnect / roaming semantics

- **WG roaming is free.** WireGuard latches onto whatever source address a valid handshake/keepalive arrives from; a network change (Wi-Fi→cellular) that keeps the carrier reachable needs **no transport re-dial** — WG roams on top [01-DP §8 step 2]. The transport only re-dials when the *carrier* dies (socket error, QUIC CONNECTION_CLOSE).
- **Per-network memory.** On a successful dial, the ladder remembers the working `TransportKind` keyed by network identity (SSID / gateway fingerprint) so a reconnect on the same hostile network skips straight to what worked [01-DP §5.3 step 4]. This is ladder state (§6), not transport state.
- **Reconnect backoff** (UNVERIFIED exact schedule — orchestrator owns it): recommended exponential backoff with jitter between full ladder

sweeps to avoid thundering-herd re-dials; the per-rung failure budget (§6.2) governs within a sweep.

5.5 Timeout & budget constants (carrier-level)

Constant	Recommended default	Meaning	Source
DIAL_TIMEOUT	5 s (UNVERIFIED, tune)	max time for dial before DialTimeout	[01-DP §3.1 "bounded dial timeout"]
silence_thresh (last_recv_age)	15 s (UNVERIFIED)	carrier→Degraded if no recv	derived from §6.3
CLOSE_GRACE	2 s (UNVERIFIED)	max time close() may take to flush	§2 close contract
per-rung FailureBudget	{ max_handshakes: 3, window: 10 s } (UNVERIFIED)	dials/handshakes before escalate	[01-DP §5.3]

All four are **calibrated on Phase-0 fixtures** (§11.4.107(13) no-hardcoded-from- literature) and recorded in the BENCH evidence, not asserted [04_P0 §8].

5.6 close() semantics

close is **idempotent** and bounded by CLOSE_GRACE. Per carrier: plain-udp drops the socket (Ok immediately); masque-h3/hysteria2 call quinn::Connection::close(0, b"bye") and let quinn flush [01-DP §3.3]; shadowsocks/udp-over-tcp send TCP FIN. After close, send/recv return Err(Closed). A second close is a no-op Ok().

6. dial() and the escalation-ladder constructor

6.1 dial — the single constructor

```
/// Build a live, CarrierUp Transport from a resolved config.  
/// Returns within DIAL_TIMEOUT (§5.5). On DialTimeout/  
/// HandshakeFailed/  
/// EndpointBlocked the LADDER (§6.4) escalates to the next rung.  
/// On Ok, the returned Box<dyn Transport> is in state CarrierUp  
/// (§5.1).
```

```

pub async fn dial(cfg: TransportConfig) -> Result<Box<dyn
    Transport>, TransportError> {
    match cfg {
        TransportConfig::PlainUdp { peer, bind }      => /* §4.1
        */,
        TransportConfig::MasqueH3 { url, sni, bind, congestion }
        => /* §4.2 */,
        TransportConfig::ConnectIp { .. }             => /* §4.5
        feature-gated */,
        TransportConfig::Shadowsocks { peer, method, psk } => /*
        §4.3 */,
        TransportConfig::UdpOverTcp { peer, tls_sni } => /* §4.4
        */,
        TransportConfig::Lwo { peer, bind, session_key } => /*
        §4.6 */,
        #[cfg(feature = "hysteria2")]
        TransportConfig::Hysteria2 { url, sni, salamander_pw }
        => /* §4.7 */,
    }
}

```

dial is the **only** way to obtain a Transport; impls expose no public new that bypasses the bounded-timeout + state-machine guarantees.

6.2 The ladder policy (owned by helix-core/src/ladder.rs, consumed here)

```

// helix-core/src/ladder.rs ([01-DP §5.3])
pub struct TransportPolicy {
    pub order: Vec<TransportKind>, // coordinator-pushed; e.g.
    regional prior (§6.5)
    pub pin: Option<TransportKind>, // user manual override –
    skip the ladder
    pub budget: FailureBudget,      // { max_handshakes: u8,
    window: Duration } per rung
}
pub struct FailureBudget { pub max_handshakes: u8, pub window:
    Duration }

// Default order (unrestricted network): [ PlainUdp ]
// Restricted prior (censored region, coordinator): [ PlainUdp,
    Lwo, MasqueH3, Shadowsocks, UdpOverTcp ]

```

6.3 Selection algorithm (the rung-walk) [01-DP §5.3]

1. Start at pin if set (manual override), else order[0].
2. Resolve the rung's TransportKind to a TransportConfig (hydrate secrets from the NetworkMap), call dial(cfg).
3. On Ok(carrier) within the rung's FailureBudget → done; the WG handshake proceeds on top; emit Connected{transport, rtt} once WG establishes.

4. On DialTimeout/HandshakeFailed/EndpointBlocked **exceeding the rung's budget** → escalate to the next rung; emit Reconnecting.
5. A Degraded carrier (§5.1) that crosses `silence_thresh` is treated as a soft failure and triggers escalation **without** waiting for a hard error (pre-emptive, reads `health()`).
6. On success, **remember the working TransportKind per-network** (SSID / gateway fingerprint) — reconnects on the same hostile network skip straight to what worked [01-DP §5.3 step 4].
7. Telemetry records **only** "transport X succeeded after N escalations in region R" (aggregate, no per-user data, I5) → Censorship-Evasion Success dashboard [01-DP §5.3 step 6].

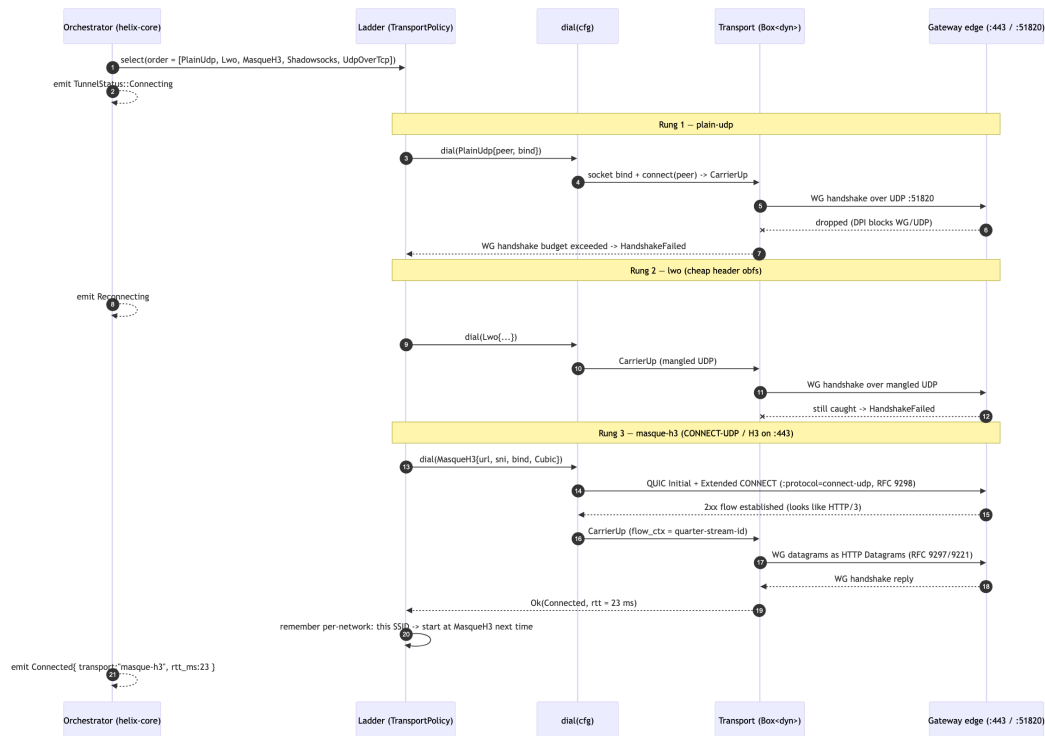
6.4 Atomic carrier swap

On escalation the orchestrator: `close()` the failed carrier (best-effort, ignore error) → dial the next rung → on Ok, atomically swap the active `Box<dyn Transport>` under the transport lock (§2.2) → resume the send/recv loops against the new carrier. The swap never happens mid-datagram (datagrams are atomic, DG).

6.5 Regional prior & per-leg topology

- The coordinator may push a **regional prior** (e.g. CN-resolved clients start at Shadowsocks) so users in censored regions don't pay escalation latency every time [01-DP §5.3 step 5].
- **Per-leg topology (decision D6)** is free: each leg is an independent Transport selection. The coordinator pushes a `TransportPolicy.order` *per leg* — user[?]gateway may start at `masque-h3/hysteria2` while gateway[?]connector stays `plain-udp` on cooperative networks. **No code change beyond per-leg policy** [01-DP §11.3, SYNTHESIS §3 D6]. **D6 SURFACED.**

6.6 Ladder sequence (the headline diagram)



7. Error taxonomy

```
// helix-transport/src/error.rs ([01-DP §3.1])
#[derive(thiserror::Error, Debug)]
pub enum TransportError {
    #[error("dial timed out")] DialTimeout,
    #[error("handshake failed: {0}")] HandshakeFailed(String), // -> ladder escalation
    #[error("peer endpoint blocked")] EndpointBlocked, // DPI/firewall verdict
    #[error("transport closed")] Closed,
    #[error("oversize datagram {0} > mtu")] Oversize(usize),
    #[error("io: {0}")] Io(#[from] std::io::Error),
    #[error("quic: {0}")] Quic(String),
}
```

7.1 Per-variant semantics & ladder coupling

Variant	Raised by	Carrier-state effect	Ladder action
DialTimeout	dial exceeding DIAL_TIMEOUT	Dialing→DialFailed	escalate to next rung
HandshakeFailed(why)		Dialing→DialFailed	

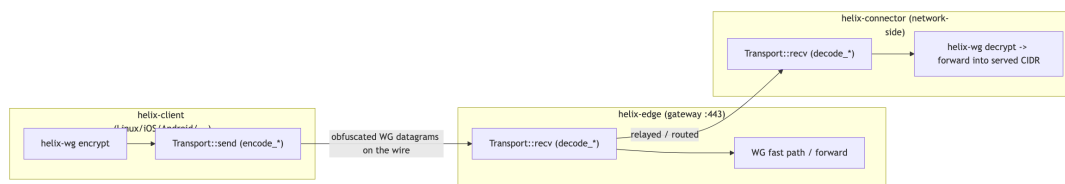
Variant	Raised by	Carrier-state effect	Ladder action
	dial (QUIC/ CONNECT-UDP failed) or orchestrator (WG handshake budget exceeded)		escalate (the canonical escalation trigger)
EndpointBlocked	send/dial on ICMP port-unreachable / TCP RST / connection-refused (DPI/firewall verdict)	CarrierUp→CarrierDown	escalate
Closed	send/recv after close or peer FIN / QUIC CONNECTION_CLOSE	→CarrierDown	orchestrator → Reconnecting/ Down no escalation; orchestrator lowers inner MTU / L3- fragments (§4.8 rule 3)
Oversize(n)	send when datagram.len() > effective_mtu + headroom	none (rejects the one datagram)	counted in send_errors; sustained → escalate
Io(e)	any syscall error	varies; repeated → Degraded	sustained → escalate
Quic(s)	quinn errors (masque/ hysteria2/connect-ip)	varies	sustained → escalate

7.2 Error classification rules (§11.4.6 no-guessing)

- An error is **never** labelled "transient/flaky" without captured evidence (§11.4.6/.50) — the RACE/SC test types (§9) deterministically reproduce each class N times.
- EndpointBlocked vs Io discrimination is **observational, not guessed**: EndpointBlocked requires a concrete signal (ICMP unreachable, RST, refused); ambiguous silence is last_recv_age Degraded, not EndpointBlocked.
- HandshakeFailed carries a String reason (e.g. "no QUIC datagram support", "CONNECT-UDP 4xx", "WG no handshake reply in budget") for the §6 telemetry — **never** a secret value (§11.4.10).

8. How the same trait serves client, connector, and edge (I4)

One trait, three consumers, N transports [04_P0 §4.2]. The byte-for-byte shared-code guarantee is the whole point of the crate [01-DP §0.1 I4].



8.1 Client role

Captures the device default route, encrypts in `helix-wg`, optionally shapes in `helix-daita`, then `Transport::send`s. Runs the **ladder** (§6) — it is the only role that *escalates* (the edge/connector accept whatever carrier the client dials). `effective_mtu()` sets the inner WG MTU [01-DP §5.1, §10].

8.2 Connector role

The **same core**, different routing posture: instead of capturing the default route it (a) advertises served CIDRs, (b) NATs/forwards decapsulated packets into the LAN [01-DP §5.1]. It is a Transport **recv-side** consumer for gateway↗connector traffic; on cooperative networks this leg typically stays `plain-udp` (per-leg D6, §6.5) [01-DP §11.3]. One binary, `--mode={client|connector}` [01-DP §5.1].

8.3 Edge role (gateway MASQUE termination — decision D5)

The edge is the **third consumer of helix-transport** [01-DP §2]. It runs the **inverse** of every `encode_*/decode_*`: `decode_http_datagram` where the client encoded, terminating CONNECT-UDP and emitting plain WG datagrams to the kernel WG fast path (§4.2). **Decision D5 (Rust vs Go edge) is SETTLED BY THE PHASE-0 G4 BENCHMARK** [01-DP §11.2, SYNTHESIS §3 D5]: this spec **assumes Rust** (shares `helix-transport` byte-for-byte, no GC on the hot path), with Go (`quic-go+masque-go`) as the conformance-suite-mitigated fallback if MASQUE-in-Rust proves painful (dual impls kept in sync by `**shared test vectors`

- a cross-edge conformance suite**) [01-DP §11.2, research-masque §4].
D5 SURFACED.

Conformance-suite contract (anti-bluff, §11.4.27/.50). If the Go edge is adopted, the I4 guarantee is preserved by a **shared encode/decode test vector corpus**: a frozen set of (WG datagram, on-wire bytes) pairs that BOTH the Rust client encode and the Go edge decode (and vice versa) must satisfy bit-for-bit. A divergence is a release blocker — it is the mechanical proof that "the client obfuscates with the same logic the edge de-obfuscates."

9. Test points — tied to the §11.4.169 closed test-type vocabulary

Every Transport workable item declares its required test types from the §11.4.169 closed set; the ONLY permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason, never a silent gap [06-phase0-spike-wbs §0.4]. Four- layer enforcement per §11.4.4(b) applies to every closure.

Code (§11.4.169)	Type	Concrete test point for helix-transport	Evidence (§11.4.5/.69/.107)
UT	unit	encode_http_datagram/ decode_* round-trip is identity for random WG- sized Bytes; effective_mtu boundary; TransportError mapping; SecretBytes Debug redaction + zeroize-on- drop	cargo test output; redaction assertion
IT	integration (real System, netns + edge + connector booted via containers submodule §11.4.76)	dial each carrier against a real edge; send→peer recv yields identical bytes; CONNECT-UDP flow really establishes	netns capture; per-carrier round- trip log
E2E	end-to-end	client overlay ns → curl http://10.10.0.20/ over each carrier, incl. plain- UDP blocked (MASQUE only)	curl hello-page capture (G1/G2)
FA	full-automation (§11.4.25/.52/.98, deterministic §11.4.50)	one-shot make spike, N=3 identical ladder- escalation runs producing	3× identical status-trace artifacts

Code (§11.4.169)	Type	Concrete test point for helix-transport	Evidence (§11.4.5/.69/.107)
		the same ordered TunnelStatus trace	
CH	Challenges (challenges submodule §11.4.27(B))	per-gate Challenge scores the captured G1/G2 evidence, not config	Challenge result.json
HQA	HelixQA (helix_qa submodule)	autonomous QA session drives dial→escalate→connected across carriers	HelixQA session evidence
LOAD	DDoS / load- flood	1/10/100 concurrent carriers + handshake- flood at dial; no fd leak, bounded mem	iperf3 + handshake-flood CSV
SEC	security (§11.4.10 + security submodule)	(a) tshark shows MASQUE flow = HTTP/3, no WG signature (G2); (b) secret- leak grep: plant a SecretBytes value, prove it appears in no log/ tracked file; (c) no- plaintext-leak: recv bytes never decrypted in-crate (I1)	tshark classification; grep-empty proof
SC	stress + chaos (§11.4.85)	iface flap, netem loss 5% (MASQUE goodput > UoT strawman [04_P0 §5.3]), process-kill mid-transfer → reconnect; close during in-flight send	goodput-under- loss CSV; recovery trace
CONC	concurrency / atomicity	concurrent send (task A) + recv (task B) on one carrier; atomic carrier- swap (§6.4) under load drops zero datagrams mid-swap	concurrency harness log
RACE	race / deadlock	cargo test --features loom on the health-cell + swap; -race on the Go edge if adopted	loom report
MEM	memory	RSS under iOS NE ceiling for plain-udp and masque	

Code (§11.4.169)	Type	Concrete test point for helix-transport	Evidence (§11.4.5/.69/.107)
		(G3 make-or-break); no per-datagram heap growth	/proc/ Instruments RSS sample
BENCH	benchmarking / performance	bench.sh: per-carrier throughput / CPU-per- Gbps / p99 incl. GC tail; effective_mtu measured not assumed	throughput/CPU/ p99 CSV

9.1 Carrier-specific acceptance gates (Phase 0) [01-DP §12.1, 04_P0 §8]

- **G1 plain-udp** — iperf3 through-tunnel $\geq$ 80% bare link + curl to LAN host succeeds (BENCH + E2E).
- **G2 masque-h3 through DPI block** — works with plain WG **blocked** (nft rule), tshark classifies HTTP/3 with **no WG signature**, goodput @ 5% netem loss **beats the UoT strawman** (SC + SEC + E2E).
- **Per-transport** — effective_mtu() **measured** (not assumed); ladder escalation captured as an **ordered TunnelStatus event trace** (FA); Oversize is a hard error, proven by a denied-oversize capture.

9.2 Anti-bluff rule for this crate (§11.4.5/.69/.107)

Every Transport PASS ships **captured wire/throughput evidence**, never a config-only or grep-only PASS. A green send/recv unit test does **not** prove a carrier works end-to-end — the IT/E2E netns round-trip with a real edge is the authoritative evidence [04_P0 §8, §12.1]. The MASQUE "looks like a browser" claim is proven by the tshark G2 capture, **not** asserted [04_P0 §5.4, §8].

10. Security considerations

#	Consideration	Mitigation in this spec	Source
S1	Plaintext never reaches the transport	I1 enforced by Bytes-opaque API; SEC test proves no in-crate decrypt; even lwo mutates ciphertext only	[01-DP §0.1 I1]
S2	Transport secrets must not leak	SecretBytes: redacted Debug, zeroize-on-drop, no Serialize,	§3.2, §11.4.10

#	Consideration	Mitigation in this spec	Source
S3	QUIC/TLS fingerprint mimicry is an OPEN problem	never logged; SEC plant-and-grep proof (§9) Mullvad does NOT publish its anti-fingerprint layer (uTLS-style ClientHello/SNI/ALPN/transport-param mimicry of a real browser). Treat as an explicit research/risk item , not a solved checkbox; sni/congestion are the seams. Do not claim DPI-proof.	[research-masque §2, §5] (honest gap)
S4	No durable per-connection state	I5: only aggregate TransportHealth counters; no per-packet/per-conn durable log; telemetry is "transport X after N escalations in region R" aggregate	[01-DP §0.1 I5, §5.3]
S5	Default-deny survives the carrier	the transport carries only what WG AllowedIPs + the edge verdict map permit; a carrier compromise does not bypass policy (defense in depth, [01-DP §7])	[01-DP §7]
S6	Double-crypto / double-CC cost is real	WG-over-MASQUE = WG AEAD inside QUIC AEAD + QUIC CC over UDP; throughput loss is acknowledged, BENCH must capture WG-direct vs WG-over-MASQUE — do not claim parity speed	[research-masque §2, §5]
S7	Masquerade decoy	edge serves a believable decoy to non-CONNECT-UDP probes; verified by SEC G2	[01-DP §3.3, 04_P0 §5.4]

11. Performance budget

Metric	Budget / target	Basis	Source
plain-udp throughput	≥ 80% of bare-link (the G1 baseline every carrier is measured against)	G1 gate	[01-DP §3.2, 04_P0 §8]

Metric	Budget / target	Basis	Source
masque-h3 throughput	≥ 50% of plain-udp (G2 bar); real loss from double-crypto/double-CC	G2 gate; acknowledged tax	[04_P0 §8, research-masque §2]
masque-h3 goodput @ 5% loss	> UDP-over-TCP strawman (the mobile-loss reason this feature exists)	G2 sub-bar	[01-DP §3.3, 04_P0 §5.3]
effective_mtu (masque)	1280 floor, measured & tuned in Phase 0 (not assumed)	MTU budget	[01-DP §10, 04_P0 §5.2]
iOS NE RSS (plain-udp + masque)	under device ceiling with ≥ 30% headroom (G3 make-or-break)	G3 gate	[04_P0 §1 G3]
health() sample cost	callable ≥ 10 Hz without contending hot send/recv	§2.3 contract	derived
CC selection	Cubic default; Bbr for lossy mobile (Hysteria "Brutal" lineage as window-ratio reference)	tuning knob	[01-DP §3.3, 04_ARCH §14]
dial latency	plain-udp sub-ms (bind only); masque-h3 = QUIC+CONNECT-UDP RTT (bounded by DIAL_TIMEOUT)	§4.1, §4.2	derived

All budgets are **captured in the Phase-0 BENCH CSV** (throughput @ 1/10/100 clients, CPU/Gbps, p99 incl. GC tail, handshake churn/sec, memory under churn) [04_P0 §7.2, §8] — measured evidence, never asserted (§11.4.6/.123).

12. Frozen-contract status & cross-doc seams

The Transport trait (§2), TransportConfig/TransportKind (§3), TransportError (§7), and dial (§6) are **frozen Phase-0 contracts** designed to survive into Phase 1 — their *implementations* may evolve, their *surface* may not without an explicit decision (§11.4.66) [04_P0 §4.2, §14, 01-DP §13].

Cross-doc seams: `TransportConfig` is **sourced** by doc 03 (`WatchNetworkMap` — the live source of the static `map.json` seam [01-DP §6.2]); `effective_mtu()` feeds `helix-wg` inner-MTU ([01-DP §4, §10]); the carrier lifecycle projects into `TunnelStatus` owned by `helix-core` [01-DP §5.2]; the edge consumer's MASQUE termination is settled by Phase-0 G4 (decision D5, [01-DP §11.2]). Out of scope here: WG crypto internals (doc 01 §4), orchestrator loops (doc 01 §5.1), platform TUN shims (doc 06).

*End of nano-detail specification — The Transport Trait (Volume 2, Data Plane). Pairs with [01-DP §3] (the overview this deepens), doc 03 (`WatchNetworkMap`, the `TransportConfig` source), and the security doc (QUIC/TLS fingerprint mimicry research item S3). Surfaced decisions: **D1** primary obfs transport (§4.7), **D5** edge language (§8.3), **D6** per-leg topology (§6.5) — all presented, none silently resolved (§11.4.66).*